

The Economic Effects of Long-Term Rent Control: The Case of New York City

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Abstract

This article presents a simple model of rent control in which the costs and benefits can be analyzed in terms of Hicksian consumer and producer surpluses. Data from New York City are then used to examine the effect of the long-term rent control in that city. Estimates are presented which suggest that the program provided tenants with benefits amounting to about 5 percent of their income in the years 1965 and 1968. Additional evidence shows that tenant benefits declined rather sharply between the two years, and that in each of the years benefits were higher for older tenants, richer tenants, and white tenants than for their counterparts. Finally, evidence presented here suggests that, due to the in-kind nature of the program, the cost to landlords exceeded the benefits to tenants by about 75 percent.

Key words: rent control, cost benefit analysis

Rent control programs are in effect in a growing number of American and Canadian towns and cities. Empirical studies using data from New York City (Olsen, 1972; DeSalvo, 1971; Roistacher, 1972; Linneman, 1987; Gyourko and Linneman, 1989), Los Angeles (Rydell et al., 1981; Fallis and Smith, 1984), Toronto (Fallis and Smith, 1985), Vancouver (Marks, 1984), and several cities in upstate New York (Vitaliano, 1986) have examined various effects of those rent control programs. Each study produced some estimate of the cost of rent control to owners of controlled rental housing. However, with the exception of Olsen (1972), the studies either treated tenant benefits as being equal to costs or failed to produce an estimate of tenant benefit.¹ The assumption that tenant benefits are equal to owner costs is a reasonable one for newly implemented rent controls. Then, benefits will differ from costs only to the extent that tenants would prefer to spend some of their rent savings on additional housing. However, as rent control programs remain in effect for a sustained period of time, the difference between costs and benefits will inevitably increase. Although it may not be important to make a distinction between benefits and costs in studies of recently implemented rent control programs, this distinction is essential in examining a rent control program like New York's where rent control has been in effect for such a long period of time.

The purpose of this study is to examine data from New York City and to determine what impact the New York City rent control program has had on the residents of controlled rental housing and on the owners of that housing. In particular, answers to the following questions are provided: (1) How large are the benefits that rent control provides to occupants of rent-controlled housing? (2) What costs do the owners of this housing incur in providing these benefits? (3) What effect does rent control have on the consumption of housing services and nonhousing goods by families in rent-controlled housing? (4) How are benefits from rent control related to the demographic characteristics of the tenant families? (5) How have these costs and benefits changed over time under rent control?

The data used to provide answers to these questions are obtained from the New York City Housing and Vacancy Surveys of 1965 and 1968. Data from later surveys are available, but they are not used here because changes in New York's rent control program after 1968 greatly reduced the number of uncontrolled rental units in that city. The methodology used here requires data for a large number of uncontrolled units.²

Although the surveys in each of the two years covered 35,000 housing units and their occupants, this study is based on observations from the rental units that are occupied, unfurnished, and privately owned. Missing data preclude the use of many observations, so the estimates generated in this study are based on samples of far less than 35,000 observations.³

1. Rent control in New York City prior to 1969

Rent control was first imposed in New York City in 1942 and was administered by the federal government's Office of Price Administration (OPA) as a part of the nationwide program of wartime wage and price controls.⁴ At the end of the war the authority to control rents was transferred to the State of New York Temporary Rent Commission where jurisdiction remained until 1962. Then, the New York State Legislature authorized the City of New York to assume responsibility of controlling rents within the city. As a consequence of different policies by the three rent control authorities, New York City developed an unusual mix of controlled and uncontrolled rental housing. Between 1943 and 1947 all privately owned rental housing units were subject to control. Then, the OPA adopted a policy of decontrolling some duplexes and single-unit rental houses and of exempting newly constructed units from control. The state continued these policies when it assumed jurisdiction, and in 1953 it provided for the decontrol of all apartments of two or fewer units as they became vacant. In 1964 the city decontrolled all units whose ceiling price exceeded \$250 per month and initiated a policy of treating substantially rehabilitated units as newly constructed units which exempted them from subsequent control. Finally, it decontrolled units that had been occupied for a year or more by the owner. The consequence of these policies was that the percentage of privately owned, renter-occupied housing units that were uncontrolled rose from 4 percent in 1950 to 33 percent in 1968.⁵

In addition to creating a substantial uncontrolled sector, New York's administration of rent control caused ceiling rents to differ widely for similar rental housing units. When controls were initially imposed in November of 1943, all rents were rolled back to their level on March 1 of that year where they remained until 1950. Subsequently, rent increases

were permitted when capital improvements were made, when additional services were provided, and when the number of tenants in a unit increased. Later a hardship provision was adopted under which rent increases were permitted if the owner of a unit could demonstrate that the prevailing ceiling rent denied him the opportunity to make a fair return on his investment. The state also permitted a 15 percent increase in the ceiling rent to go into effect automatically if a new lease of two or more years' duration was offered to a tenant, provided that this increase was acceptable to the tenant. In 1953 the state also allowed an increase in all ceiling rents to a level 15 percent above the base rent of 1943. At the same time, it provided for rent decreases in cases where there was a reduction in the level of provided services.

Under these regulations the average monthly ceiling rent increased from \$72 in 1960 to \$78 in 1965 and \$88 in 1968, but the rate of increase for individual units varied widely.⁶ Since the ceiling rent on a particular unit depended on many factors other than the quality of the unit, it is highly misleading to view the rent control program in New York as one that reduces rents on controlled units by some uniform percentage or some fixed dollar amount below the free market price.

2. The model

The model used here to examine the effects of rent control is identical to the model used by Olsen-Barton (1982) to examine public housing. Their model was adapted from one initially developed by DeSalvo (1971). In this model the behavior of a typical consumer is analyzed under the following assumptions:

1. There are two composite goods, housing services (H) and nonhousing goods (X).
2. The uncontrolled market for both housing services and nonhousing goods is perfectly competitive and in long-run equilibrium.
3. The long-run supply curve in each uncontrolled market is perfectly elastic.⁷
4. Information is costless.
5. The supply price of housing in the uncontrolled market is not affected by the existence of rent control.
6. Housing services are produced as efficiently in the controlled housing market as they are in the uncontrolled housing market.

Assumptions 2 through 5 imply that the prices currently in effect in the uncontrolled markets for housing and nonhousing would be unaffected in the long run if rent control were ended. Assumption 6 is made to facilitate the estimation of owner costs.

Under the above assumptions, the situation of a typical family under rent control and decontrol can be expressed in terms of figure 1. With an income equal to Y_1 and market prices P_h^m and P_x for housing and nonhousing, in the absence of rent control this family would maximize utility by selecting consumption point m . If a program of rent control were implemented, this family might have the opportunity to obtain some consumption point,

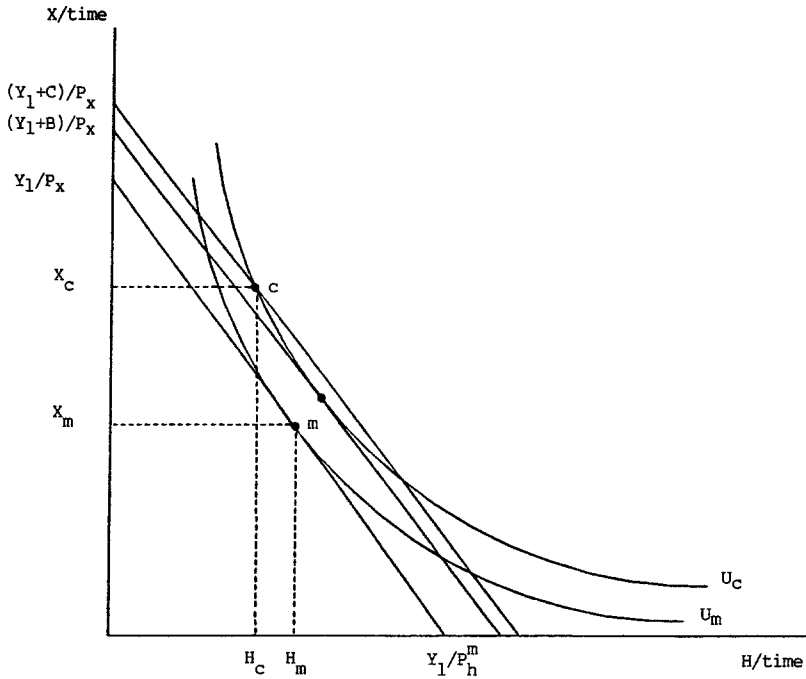


Fig. 1. The cost and benefit of rent control.

like point c which is beyond the original budget line. An effective rent control program will set the ceiling rent of a controlled unit

$$R^c = P_h^c H \quad (1)$$

below the market rent of that unit

$$R^m = P_h^m H. \quad (2)$$

The existence of such an opportunity depends upon whether this particular family can locate a vacant controlled unit. Because of the low vacancy rate in the controlled sector, most families cannot.

The exact location of point c depends upon the vacant unit found (the number of units of housing service emitted by that unit) and on the relationship between the ceiling rent of that unit and its market value. Given the way ceiling rents are established in New York's rent control program, the relationship between R^c and R^m differs from unit to unit. The only thing that can be said with certainty about point c is that it lies above the original budget line if the ceiling rent of that unit is an effective one, and that it lies below a horizontal line

at Y_1/P_x because ceiling rents are positive. If point c lies beyond the original indifference curve, the family will move into the controlled sector; and if more than one controlled unit is made available to this family, their choice will depend on which unit enables them to attain the highest utility.

In the absence of rent control, this family would require an income of at least $Y_1 + B$ to afford a combination of housing and nonhousing that is equally satisfying to that attained under rent control. The magnitude of B is determined by a parallel outward shift of the original budget line until it is tangent to the indifference curve that passes through point c .

In terms of figure 1, this family receives benefits from rent control equal to B dollars. This measures the increase in consumer surplus in terms of the Hicksian equivalent price variation.

Under rent control, this family consumes $Y_1 + C$ dollars worth of goods with an income of Y_1 . Since they pay market prices for nonhousing goods, C is equal to $R^m - R^c$, the difference between the market value of their unit and its ceiling price. Assuming that it is equally costly to provide housing services in the controlled and uncontrolled housing markets, C represents the owner's reduction in net rental income as a result of rent control; and it is used here to measure the owner's annual cost.

The difference between H_c and H_m and between X_c and X_m shows the impact of rent control on housing consumption and nonhousing consumption, respectively, for this particular family.

3. Estimating procedures

To make the model estimable, it is assumed that family preferences can be expressed by

$$U = (H - B_h)^a (X - B_x)^{1-a} \quad (3)$$

where H and X are defined as before and a , B_h , and B_x are parameters.⁸ Maximizing this displaced Cobb-Douglas utility function subject to the budget constraint

$$Y = P_h^m H + P_x X \quad (4)$$

yields the following demand relationships:

$$H = B_h + (a/P_h)(Y - P_h B_h - P_x B_x), \quad (5)$$

and

$$X = a B_x + ((1-a)/P_x)(Y - P_h B_h - P_x B_x). \quad (6)$$

Housing expenditure can be expressed as

$$R^m = P_h H = (1-a)P_h B_h + aY - aP_x B_x, \quad (7)$$

and nonhousing expenditure as

$$P_x X = (1-a)(Y - P_h B_h) + a P_x B_x. \quad (8)$$

Since equations 5 and 6 can be used to predict housing and nonhousing consumption at various income levels, they can be used to predict consumption patterns if rent control were ended.

To determine the size of a family's benefit from rent control, desired quantities of housing and nonhousing are expressed in terms of equations 5 and 6 with income equal to $Y_1 + B$.

$$H = B_h + (a/P_h)((Y+B) - P_h b_h - P_x B_x) \quad (9)$$

and

$$X = a B_x + (1-a)/P_x((Y_1 + B) - P_h B_h). \quad (10)$$

Expressions 9 and 10 can be substituted into the utility function to determine the level of satisfaction associated with this consumption point.

$$U = (B_h + (a/P_h)((Y+B) - P_h B_h - P_x B_x) - B_h)^a (a B_x + ((1-a)P_x)((Y_1 + B) - P_h B_h - B_x)^{1-a}. \quad (11)$$

Since the level of utility associated with that consumption point is equal to that associated with consumption point c ,

$$U_b = U_c = (H_c - B_h)^a (X_c - B_x)^{1-a}. \quad (12)$$

Setting 11 equal to 12 and solving for B yields

$$B = ((P_h H_c - P_h B_h)/a)^a ((P_x X_c - P_x B_x)/(1-a))^{1-a} + P_h B_h + P_x B_x - Y_1. \quad (13)$$

Finally, the cost of the program can be expressed as

$$C = R^m - R^c = P_h^m H_c - P_h^c H_c. \quad (14)$$

Examination of equations 5, 6, 13, and 14 reveals that the benefit of rent control to a particular family, the cost of that program to the owner of the controlled unit, and the consumption patterns of this particular family under rent control and decontrol can be expressed in terms of the following variables:

1. Family income, Y_1 .
2. The ceiling rent of the controlled unit, R^c .
3. The market price of housing and nonhousing, P_h and P_x .
4. The market value of the controlled unit, $P_h^m H_c$.
5. The family's indifference map parameters, a , B_h , and B_x .

The first two variables are obtained from the sample data, but the others must be obtained by other means. It would be possible to pool the data for the two years and to use price data from the Bureau of Labor Statistics' indices for both housing and other goods. However, a majority of the housing units in that sample were subject to rent control, so these indices report changes in relative prices that do not accurately measure market prices.⁹ In addition, the change in prices between these two years was too small to produce reliable estimates of the effect of price level changes. This problem is handled by treating data for each of the two years separately. For each year units of both housing service and nonhousing goods are defined such that the unit price of each is one dollar.

The market value of each controlled unit is estimated through the use of a hedonic equation. For each of the two years, all unfurnished, uncontrolled, and occupied units in the sample are selected. The uncontrolled rent is regressed on those housing characteristics which are reported in the data set and on some demographic characteristics of the tenants of those units. Since there is no theoretically proper functional form for a hedonic equation, the linear form is used here. Specifically, the equation estimated for each year is

$$R = a \sum_{i=1}^{18} B_i X_i \quad (15)$$

$$\text{Where } R = \frac{\text{RENT} + \text{UTILITIES}}{\text{ROOMS}}$$

and the X_i 's are as identified in table 1.¹⁰

Two variables that do not directly affect levels of housing services—race and income—are included in this equation. Income is included as a proxy for neighborhood quality for which there is no information in the data set other than the borough where the unit is located. The race dummy also serves this purpose and, in addition, accounts for the possibility of racial discrimination in the rental housing market.

In this particular specification, units on or above floor seven are treated the same whether or not an elevator is present in the building. This does not create a problem since every unit of this type in the sample is in a building with an elevator.

Assuming that gross rent per room is a linear function of the regressors listed in table 1 and an error term that is independent of these regressors and that the mean market value of *omitted* housing characteristics is the same for controlled and uncontrolled units, an unbiased predictor of the annual gross market value of a controlled housing unit (R_m) can be expressed as

$$R_m = [\hat{a}_0 + \sum_{i=1}^{18} \hat{B}_i X_i] \times \text{ROOMS} \quad (16)$$

where $\hat{a}$ and $\hat{B}_i$ are the least-squares estimates of the regression coefficients in each of the two years and the X_i are the housing and demographic characteristics of the controlled units and their occupants. These coefficients are reported in table 2. Most of the coefficients are statistically significant and of the expected sign. They indicate that market rents are higher for newer units, better-conditioned units, and for units in Manhattan. In buildings with an

Table 1. Description of regression in market rent equations

Regressor	Description
X^0	Constant
X^1	1 if dwelling built after 1959; 0 otherwise
X^2	1 if dwelling built 1947–1959; 0 otherwise
$(X^1 = X^2 = 0$ if dwelling built prior to 1947)	
X^3	1 if condition of unit is sound; 0 otherwise
X^4	1 if condition of unit is deteriorating; 0 otherwise
$(X^3 = X^4 = 0$ if condition is dilapidated)	
X^5	story of unit less than 7; 0 otherwise
X^6	1 if owner resides in building; 0 otherwise
X^7	1 if story is greater than 6; 0 otherwise
X^8	$X^5 \cdot \text{ELEV}$ where ELEV is 1 if building has an elevator; 0 otherwise
X^9	1 if dwelling located in Bronx; 0 otherwise
X^{11}	1 if dwelling located in Richmond; 0 otherwise
X^{12}	1 if dwelling located in Queens; 0 otherwise
$(X^9 = X^{10} = X^{11} = X^{12} = 0$ if dwelling located in Manhattan)	
X^{13}	Inverse of the number of rooms
X^{14}	(Bedrooms/Rooms)*100
X^{15}	1 if household head is nonwhite; 0 otherwise
X^{16}	1 if rental unit is single-unit or two-unit; 0 otherwise
X^{17}	1 if utilities are provided by landlord; 0 otherwise
X^{18}	Family income

elevator, units on higher floors have higher market rent; the reverse is the case in buildings without an elevator. The coefficient of the income variable is positive in both of the years which is consistent with the notion that tenant income is positively correlated with unobserved neighborhood amenities.

The negative coefficients for both the race variable and for the variable indicating the presence of a landlord are surprising. To the extent that nonwhites were discriminated against, one would expect higher market rents for units occupied by minorities. Perhaps the race dummy is negatively correlated with unobserved neighborhood amenities with this effect swamping the effect of discrimination.

We expected to find that units with a landlord present would be provided with better service and would therefore bring a rent premium. These units turn out to have a lower market value, perhaps because tenants dislike being monitored by landlords.

The coefficient of determination is quite high in each of the years, suggesting that this set of variables does a good job of explaining the variation in market rents.

The indifference map parameters are obtained by rewriting equation 7 as

$$H/Y = a + c(1/Y) \quad (17)$$

where

$$c = (1-a)B_h - aB_x. \quad (18)$$

Table 2. Estimated relationships between market rent and housing characteristics

Regressors	Coefficients 1965	Coefficients 1968
X^0	208.3 (5.806)	134.3 (3.258)
X^1	106.7 (13.305)	106.1 (11.162)
X^2	32.2 (3.789)	42.8 (4.273)
X^3	11.0 (0.355)	77.7 (2.210)
X^4	7.3 (0.226)	98.2 (2.632)
X^5	-9.5 (-2.069)	-34.5 (-7.020)
X^6	-34.5 (-4.383)	-28.0 (-2.760)
X^7	80.5 (4.464)	88.5 (7.429)
X^8	17.6 (4.258)	38.5 (8.904)
X^9	-229.0 (-28.225)	-289.4 (-25.065)
X^{10}	-235.3 (-31.060)	-277.6 (-28.872)
X^{11}	-230.1 (-13.676)	-310.8 (-17.546)
X^{12}	-221.7 (-31.306)	-263.1 (-30.324)
X^{13}	1141.6 (52.806)	1292.9 (47.544)
X^{14}	0.437 (2.223)	1.421 (5.452)
X^{15}	-67.1 (-8.775)	-67.9 (-7.714)
X^{16}	17.4 (0.750)	27.9 (2.704)
X^{17}	-55.0 (-8.968)	-21.0 (-2.756)
X^{18}	0.010 (25.847)	0.0096 (22.896)
<i>F</i> value	781.905	611.820
R_2 -adj	.779	.731
Sample size	3994	4029
<i>T</i> -scores in parentheses		

Since occupation of rent controlled housing may distort consumption patterns, data on families in uncontrolled housing are used to estimate these parameters. Since consumption patterns are likely to be affected by demographics, families are classified by the household head's age, race, marital status, and by the family's size. Within each family class, families are assumed to have the same indifference map parameters, but these parameters are allowed to differ for families of different types.

Assuming an error term independent of $(1/Y)$, ordinary least-squares estimates of equation 17 will produce unbiased estimates of the indifference map parameters for families of each type. To obtain separate estimates of each of the two displacement parameters, either B_h or B_x must be estimated independently. The displaced Cobb-Douglas utility function assumed in this study implies that the level of utility for each family depends upon quantities of housing services and nonhousing goods in excess of some minimum levels which can be interpreted as subsistence levels. Therefore, the minimum level of housing expenditure for families of each type is used as an estimate of B_h . The indifference map parameter estimates are reported in table 3. These parameters imply marginal propensities to consume housing services of between zero and one for all family types, but the propensities to consume do not appear to vary systematically with the family characteristics.

4. Empirical results

By applying the parameter estimates for the hedonic rent equations and for the indifference maps to the expressions for cost, benefits, and consumption patterns, estimates of the various aggregate effects of rent control can be computed.¹¹ These computations are done both in terms of current prices and in terms of constant 1987 prices and are reported in table 4.

In terms of 1987 dollars, the mean benefit to tenants of controlled rental housing was \$1,384.41 in 1965 and \$782.64 in 1968. This represents a mean increase in real income of 6.6 percent in 1965 and 3.7 percent in 1968. The mean costs to owners of providing those benefits was \$1,937.89 and \$1554.27. Aggregating over the 1,336,000 families in controlled rental housing in 1965, total benefits were \$1.85 billion and total costs were \$2.59 billion. By 1968, the number of families in controlled units had fallen to 1,267,000. The total benefits were \$991.6 million and the total costs were \$1.97 billion. The cost to owners of providing each dollar in benefits was \$1.40 in 1965 and \$1.99 in 1968.¹²

The estimates reported here suggest that in each of the two years, rent control caused the mean family in controlled housing to consume slightly fewer housing services than they would in the absence of rent control and to consume about 10 percent more nonhousing goods. If the program had the same effect on each family as it did on the mean family, the efficiency loss from rent control would be quite small. Then, the program would have roughly the effect of extracting lump sums from landlords and transferring the cash to tenants with the restriction that the cash not be spent on additional housing. In fact, the distortions in consumption patterns for individual families are quite large. Our estimates suggest that approximately 30 percent of the families in controlled rental housing have their housing consumption cut by rent control by at least 25 percent. An additional 25 percent

Table 3. Estimates of parameters of indifference maps 1965

Race	Marital Status	Age of Household Head	Family Size	$a(h)$	B_h	B_x	R^2	Sample Size
White	M	-30	1-3	.0711	684	-6,535	.55	545
		-30	4+	.1020	444	-4,807	.40	177
		31-50	1-3	.0883	620	-5,410	.45	549
		31-50	4+	.0956	553	-5,108	.37	540
		51+	1-3	.1031	624	-4,727	.64	555
		51+	4+	.0421	780	-18,637	.61	107
	S	-30	1-2	.0678	900	-10,517	.62	236
		-30	3+	.1952	1140	212	.22	9
		31-50	1-2	.1128	720	-3,309	.52	405
		31-50	3+	.1638	1020	-422	.49	35
		51+	1-2	.1784	240	-2,482	.55	305
		51+	3+	.0938	1212	-1,372	.77	16
Nonwhite	M	-40	1-3	.0605	804	-8,490	.62	108
		-40	4+	.0815	300	-7,748	.51	78
		41+	1-3	.0694	696	-4,933	.64	52
		41+	4+	.0418	504	-18,064	.72	50
	S	-40	1-2	.1489	612	-1,398	.60	56
		-40	4+	.0824	684	-5,394	.67	39
		41+	1-2	.1612	780	-662	.66	44
		41+	3+	.1032	792	-1501	.57	22
White	M	-30	1-3	.0469	720	-14,665	.55	541
		-30	4+	.0846	912	-2,851	.42	177
		31-50	1-3	.0955	744	-3,645	.49	474
		31-50	4+	.0743	384	-12,026	.51	451
		51+	1-3	.1225	468	-4,362	.58	583
		51+	4+	.0539	480	-19,126	.60	85
	S	-30	1-2	.1138	924	-3,428	.66	232
		-30	3+	.1895	1044	75	.21	17
		31-50	1-2	.0963	696	-5,213	.51	310
		31-50	3+	.1291	816	-1707	.55	37
		51+	1-2	.1675	204	-3,276	.54	372
		51+	3+	.0125	1104	-60,302	.50	18
Nonwhite	M	-40	1-3	.0530	627	-10,144	.60	106
		-40	4+	.0849	576	-5,499	.47	117
		41+	1-3	.1249	612	-1,397	.58	74
		41+	4+	.1116	360	-5,917	.60	105
	S	-40	1-2	.1024	360	-5,917	.60	105
		-40	4+	.0728	684	-2,617	.82	37
		41+	1-2	.1233	384	-3,301	.36	62
		41+	3+	.0655	924	-5,519	.78	53

Table 4. Aggregate effect of rent control

	1965		1968	
	Current Dollars	1987 Dollars	Current Dollars	1987 Dollars
Mean benefit	\$ 374.57	\$ 1,387.41	\$ 232.93	\$ 782.64
Mean cost	524.32	1,937.89	462.58	1,554.27
Mean <i>H</i> under rent control	1,524.40	5,634.18	1,587.04	5,332.45
Mean <i>X</i> under rent control	4,695.10	17,353.90	5,107.99	17,162.84
Mean predicted <i>H</i> under decontrol	1,550.75	5,731.57	1,625.65	5,478.80
Mean predicted <i>X</i> under decontrol	4,144.43	15,317.81	4,606.80	15,478.85
Mean percent change in <i>H</i> under decontrol	1.7%		2.4%	
Mean percent change in <i>X</i> under control	-11.7%		-9.9%	
Sample size	5395		3961	

of the families consume at least 25 percent more housing under rent control than they would in its absence. Only 15 percent of the families consume within 10 percent of the quantity of housing services under rent control as they would in the absence.

In evaluating a subsidy program, in addition to considering the relative size of costs and benefits, it is useful to know how they are distributed. The Housing and Vacancy Surveys provide no information about the owners of controlled units, so no attempt is made to examine the distribution of costs. Even if this information were available, it would be of questionable value since a portion of the cost of the program is in the form of capital losses to prior owners who sold controlled units after rent controls were adopted. There is no reason why owners who purchased controlled units after the enactment of rent control would fail to realize competitive returns, so long as the program continued to operate as expected.

The surveys do provide information about the age, race, income, and family size of tenant families, so it is possible to examine the relationship between tenant benefits and these demographic characteristics. Since rent control is designed to assist tenants of private rental units, the distribution of benefits is examined both within the sample of families residing in controlled rental units and within the sample of all families residing in private rental housing. Table 5 provides a profile of tenants in the controlled and uncontrolled sectors. In each year there is a higher proportion of minority families in the controlled sector, and the families in that sector are older and poorer than their counterparts. Family size is essentially equal in the two sectors.

To determine the importance of demographic characteristics on the distribution of tenant benefits, net benefit is regressed on the age, race, family size, and income of tenant families. Separate regressions are estimated for families residing in rent controlled units and for all families who reside in private rental housing with a net benefit of zero assigned to families who do not reside in rent controlled housing. The regression results are reported in table 6.

Table 5. Characteristics of families in uncontrolled and controlled housing

	1965		1968	
	Uncontrolled	Controlled	Uncontrolled	Controlled
Income	\$9,832	\$5,695	\$10,342	\$6,232
Age	42.18	49.02	42.49	58.75
Family size	2.47	2.72	2.72	2.58
% Minority	13.60	26.40	15.50	32.20
Sample size	4149	5395	4072	3961

Table 6. Distribution of tenant benefits

	1965		1968	
	Control Families	All Families	Control Families	All Families
Income	0.014 (4.18)	-0.011 (-6.61)	0.164 (3.70)	-0.005 (-2.63)
(Income) ²	-1.41E-07 (-1.60)	1.23E-07 (3.27)	-4.56E-07 (-4.53)	-9.23E-09 (-0.22)
Age of household head	21.97 (7.35)	16.24 (8.70)	24.32 (5.66)	13.25 (5.88)
Age of household head squared	-0.15 (-5.23)	-0.10 (-5.42)	-0.18 (-4.12)	-0.09 (-3.86)
Family size	-12.14 (-0.74)	5.14 (0.52)	-159.28 (-6.18)	-78.07 (-6.19)
Family size squared	-0.83 (-0.41)	-1.52 (-1.21)	14.78 (4.34)	7.18 (4.44)
Race dummy (nonwhite=1)	-71.16 (-3.66)	-14.68 (-1.15)	-179.88 (-6.50)	-97.37 (-6.39)
Constant	-296.81	-213.74	-215.55	-70.97
R ² -adj	0.057	0.059	0.076	0.482
Sample size	5,395	9544	3961	8033

Note: *T*-statistics in parentheses.

In all cases the coefficient of determination is very low, indicating that the program of rent control in New York City did a poor job of providing equal benefits to similarly situated families. Among families in controlled rental housing in 1965 and in 1968, benefits are higher for wealthier and older families and lower for larger families and minority families. Since rent regulations in New York City during that time period permitted rent increases when a unit is vacated, it is not surprising to find larger benefits associated with older tenants who are less mobile. Mobility may also explain the negative coefficient for the race variable in each of the years. The proportion of minority families in controlled rental housing in New York City increased rapidly during the 1960s, causing minority tenants to have shorter tenure in their current unit. Since tenant benefits are positively

related to tenure, the smaller benefits for nonwhite families may be caused by their more recent arrival.

These results contrast sharply with those produced by Olsen (1972) using the same 1968 data set. His estimates of benefits, costs, and effects on consumption patterns are quite similar to our 1968 estimates. However, in examining the distributional pattern of benefits, his empirical findings are similar to ours only in that tenant benefits are found to be significantly higher for older families. His estimated coefficients of both the income and the race variable have signs opposite to the ones reported here. Although there are many differences in the two studies that have the potential to lead to conflicting results, we suspect that the primary one is our inclusion of demographic variables in the market rent equation. In contrast to Olsen, we elected to include income and a dummy variable for race in our market rent equation to serve as proxies for unobserved neighborhood characteristics. We did so because we found that in absence of the two variables, estimates of net benefits for high income families were implausibly low (on average they were negative), while estimated benefits for minority families were implausibly high. It was clear that this was the consequence of a predicting equation that systematically underpredicted the value of controlled rental units occupied by wealthy tenants and that overpredicted the value of units occupied by minority tenants.

The empirical results reported here for the year 1968 also differ in some ways from those of Gyourko and Linneman (1989) using the same data set. As noted earlier, Gyourko and Linneman estimated tenant benefits under the assumption that rent control does not distort consumption patterns. Given this assumption, their measure of tenant benefit is conceptually equivalent to our measure of cost. However, our estimate of mean cost is only about 60 percent as large as their mean benefit estimate. Despite differences between the two measures of tenant benefits, our findings concerning the distribution of benefits are remarkably similar to those of Gyourko and Linneman—at least for the sample of families who reside in rent controlled units. Like them, we find that tenant benefits increase with the income and age of the household head and that white families receive larger benefits than do similar minority families.

In examining the distributional pattern of benefits within the sample of all tenants in private rental housing, the coefficient of income is negative for both 1965 and 1968. One reason is that lower income families are over represented in the controlled sector, perhaps because the lower quality housing provided there is more appealing to poor tenants than to richer ones.

Between 1965 and 1968, the mean cost of rent control fell from \$1,937.89 to \$1,554.27, indicating that controlled housing prices rose more rapidly than uncontrolled housing prices. This finding is somewhat surprising, but there are at least three plausible explanations. In 1965 the vacancy rate in apartments was 3.19 percent—a very high level for that city.¹³ During the next three years, 20 percent of the tenants in controlled housing moved and 30 percent of those in uncontrolled housing moved. Since the extent of price changes in the controlled sector depends primarily on the rate with which units are vacated, one would expect large price increases for controlled housing during that period. In contrast, the high vacancy rate would tend to depress rental prices in the uncontrolled sector, making it entirely possible for average price increases there to be lower. If this is the correct explanation, there is no reason to have expected this trend to continue.

An alternative explanation of the decline in owner cost is that the liberalized rent control policies implemented during the 1950s and early 1960s represented a move toward gradual decontrol. This view of New York's rent control program would imply that declines in owner cost are normal during periods of moderate inflation.

A final explanation of the decline in owner cost is that rent control leads to less maintenance and more rapid depreciation of controlled rental units. Since prices are not permitted to adjust to clear the market, quality adjustment will tend to perform this task, ultimately causing the market value to fall to the ceiling rent with the owner's cost falling to zero.

The decline in owner cost between 1968 and 1986 was accompanied by a decline in a mean tenant benefits of \$601.77. Nearly two-thirds of this decline is attributable to the rise in controlled rents relative to market rents. The remaining one-third represents an increase in the distortion of rent control on consumption patterns between 1965 and 1968.

In generating these estimates of the costs and benefits, certain effects of rent control have been ignored. There are administrative costs to the program that both owners and tenants bear in the form of higher taxes. Owners incur additional administrative costs in complying with the rent control laws. In addition, no attempt has been made to estimate the additional search costs that tenants incur as a consequence of rent control. Had these effects been taken into account, the cost estimates would be higher; and the benefit estimate would be lower.

These estimates have also been based on the assumption that tenants in controlled rental units perform the same amount of self-maintenance as tenants in uncontrolled units. To the extent that they are induced to perform more maintenance, the benefit estimates reported here are biased upward. Our analysis has also ignored costs associated with any distorting effects that rent control might have on tenant mobility or on the decision to own or to rent.¹⁴

Finally, the analysis has been conducted under the assumption that ceiling rents are enforced. Although it is widely known that side-payments are made by tenants to obtain vacant units in the controlled sector, there is no evidence on the frequency or magnitude of these payments. To the extent that these payments occur, costs and benefits are reduced by an amount equal to the payments.

5. Summary

The empirical results reported in this study suggest that in 1965 and 1968 the New York City rent control program succeeded in its implicit objective of providing benefits to occupants of private rental housing. However, it did so in an inefficient and inequitable manner. Due to the in-kind nature of the tenant subsidies, the total cost of the program in the two years was nearly 75 percent greater than the total benefits. Inequalities resulted from the failure of the program to provide equal benefits to similar families in controlled rental housing, and from its inability to provide any benefits to families who failed to obtain a rent-controlled unit.

Although this study presents no evidence on the relationship between the cost of rent control to owners and the characteristics of those owners, it is surely the case that the costs of this program were not borne in an equitable manner. For families of any level of wealth

or income, there is certain to be an enormous variation in the extent of rental property holdings. Therefore, a program for which costs are related to rental property ownership cannot succeed in imposing equal costs on similar families.

Notes

1. The study by Gyourko and Linneman (1989) is based on the simplifying assumption that rent control does not distort patterns of housing consumption. Under this assumption, the cost of rent control to owners (the difference between the market value of a controlled rental unit and its ceiling price) is identical to the tenant benefit.

2. In a recent paper Linneman (1987) handles this problem by treating units that are subject to rent stabilization as uncontrolled. His rationale for doing so is that the two sectors do not differ with respect to either vacancy rates or mean tenant duration.

3. Furnished units were deleted from the analysis because they comprised less than 4 percent of the uncontrolled sample and because the value of furnishings is likely to vary widely from unit to unit.

4. This historical summary is based on information from Paul L. Niebanck (1970, PP. 20–22) and Elizabeth A. Roistacher (1972, pp. 13–27).

5. Niebanck (1970, p. 19).

6. Niebanck (1970, p. 54).

7. This assumption implies that the controlled sector will disappear in the long run. New York City's experience is consistent with this implication. Niebanck (1970) reports that between 1960 and 1968 alone, the number of controlled rental units fell by about 209,000, and that less than 60,000 of these units were decontrolled. Most of the others were demolished.

8. While it may be the case that alternative utility functions fit the data better, the displaced Cobb-Douglas specification is used here because it allows a direct calculation of consumer benefit. In addition, its parameters can be estimated in the absence of relative price information; and it is less restrictive than the CES or Cobb-Douglas specifications which do not permit both income and price elasticities to vary with income and price.

9. Olsen and Barton (1982, pp. 12–13).

10. Hedonic equations can be quite sensitive to the functional forms used. Therefore, we tried an alternative log specification but found that the predicted rents which that model generated differed from those of the linear model by only 2 percent.

11. Tenant benefits cannot be calculated in cases where predicted market value is less than B_h . There were 33 of these cases in 1965 and 283 in 1968, and these observations were dropped from the analysis.

12. The magnitude of this change may seem surprisingly large. However, for reasons that are discussed later, between 1965 and 1968 rents in the controlled sector rose more rapidly than those in the uncontrolled sector. Since increases in ceiling rents cause both costs and benefits to fall by equal dollar amounts, such increases will cause the ratio of costs to benefits to rise.

13. Niebanck (1970, p. 185).

14. Gyourko and Linneman (1989) provide convincing evidence that rent control in New York City reduces mobility and distorts home ownership propensities. They do not attempt to quantify the magnitude of such cost.

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